Brain Signal Processing: where we are

Detecting brain and mental disorders from resting-state EEG with graph neural networks · team update and full work history · 13 November 2026 · supervisor Dr. Nileshchandra Pikle

In one paragraph. We built and honestly tested about 30 models on 12 public EEG datasets and froze a final system per disease: 5 complementary models voting, trained on all hospitals of a disease together, with calibrated probabilities and an 'unsure' option. Final scores: Alzheimer's 0.94, FTD 0.89, depression 0.95-0.97 (one dataset), schizophrenia 0.67-0.84 depending on hospital, Alzheimer's vs FTD 0.71-0.86. The main lesson: **a bigger or fancier model does not help** - simple logistic regression ties our graph networks and transformers; better preprocessing and combining models helped more. What limits us is the data: small datasets, hospital 'fingerprints', and recording differences between patient and control groups in two datasets (found and handled). We also showed that several published high scores come from flawed testing.

0.94	0.89	0.95	0.67-0.84	0.71 / 0.86
Alzheimer's vs healthy (Thessaloniki)	FTD vs healthy (Thessaloniki)	Depression vs healthy (Mumtaz, 1-25 Hz)	Schizophrenia vs healthy (3 hospitals)	Alzheimer's vs FTD (Thessaloniki / BrainLat)

Numbers are AUC (see 'How to read the numbers'). Each comes from people the model never saw during training.

How to read the numbers

Term	Plain meaning
AUC	Chance that the model ranks a random patient as 'more ill' than a random healthy person. 0.5 = coin flip, 0.7 = useful, 0.8 = good, 0.9+ = very good.
Subject-wise testing	A person's recording is either all in training or all in testing, never split. Splitting one person's 10-second windows across both (common in papers) inflates scores, because the model just recognises the person.
Leakage	Any way the test answers sneak into training or model choice, e.g. picking the best training epoch by looking at test accuracy. Gives scores that do not hold up.
Hospital effect	Each hospital's equipment and procedure leave a fingerprint. Our models could tell the hospital from healthy people alone with AUC 0.88-1.00 - easier than the disease.
Ensemble	Several models vote (we average their rankings). Members are fixed before seeing results.
Master model	One model per disease trained on people from all hospitals mixed together, tested on held-out people from every hospital.
95% CI	Range the true score probably lies in. With 30-100 people per dataset these ranges are wide (often ±0.10), so small differences between models mean nothing.

Current scope

We narrowed to the four disorders with the best and most defensible data: **Alzheimer's (AD), frontotemporal dementia (FTD), depression and schizophrenia**. Parkinson's was explored (4 datasets, AUC 0.73-0.80) and parked to keep focus.

Disorder	Datasets used	People	Notes
AD, FTD	Thessaloniki (OpenNeuro ds004504)	36 AD, 23 FTD, 29 healthy	Clean; one hospital
	BrainLat (Argentina + Chile, Synapse)	AD, FTD, healthy (86 usable)	Patients and healthy people were recorded into different file types - see caveats
Depression	Mumtaz (Malaysia, figshare)	30 depressed, 28 healthy	Clinical diagnosis; now our only depression set
Schizophrenia	ASEEG, Warsaw, Semmelweis	101 / 28 / 71	Three hospitals

Dropped datasets

Dataset	Why dropped
UNM depression (ds003478)	'Depressed' meant a questionnaire score ≥13. Of 46 such students only 11 had depression confirmed by clinical interview; 9 were interviewed and did not meet criteria. Label noise capped scores at ~0.6-0.7 whatever we did.
ASZED schizophrenia	Eyes-open and only ~20 s per person (other sets are eyes-closed, minutes long).
14 Chilean BrainLat controls	Their signal files do not exist on Synapse (header files only).
Parkinson's sets (4)	Out of focus, kept in the repo.

Shared processing pipeline

Every recording goes through the same steps: 19 standard scalp electrodes → mains-noise filter → resample to 128 Hz → keep 1-45 Hz → average reference → cut into 10-second windows → throw away windows with extreme or flat signal → skip the first minute when the recording is long enough → keep up to 60 clean windows spread across the whole recording. From each window we compute power in 6 frequency bands per electrode and how strongly each pair of electrodes moves together (coherence), plus band power with the background 1/f slope removed. These become the nodes and edges of the brain graph.

Work history, in order

1. Proposal and baseline

- Rejected the suggested RL-based paper (IEEE 10977776) as a baseline - not a GNN-on-EEG method.
- Reproduced **EEG-GCNN** (released checkpoints, 478 held-out people) as our reference GNN.

2. Literature plan

- Reviewed 6 papers. Chose **MSBP-GNN** as the main model to rebuild, **Tigga et al.** as an example of flawed testing, and the GATv2→LSTM idea (spatio-temporal) as an extension.

3. Data and first models

- Built the raw-EEG pipeline and collected 12 public datasets.
- Trained GATv2 and GCN per disorder, rebuilt MSBP-GNN, built GATv2-LSTM. All landed in the same range; the LSTM added nothing.

4. Cross-hospital attempt (paused)

- Tried three standard fixes for hospital differences: ComBat, Euclidean alignment, domain-adversarial training (DANN). Transfer to an unseen hospital stayed near chance (e.g. schizophrenia).
- Site probe: hospital identifiable at AUC 0.88-1.00 from healthy people alone. Paused, then dropped.

5. Pushing the scores up honestly

• Added richer EEG markers (1/f slope, alpha peak, entropy, Hjorth), a Riemannian model, the pretrained LaBraM transformer, and more datasets.

• Built **Ensemble v2** (6 models, fixed in advance) - our headline within-dataset numbers.

• Target '0.85 everywhere' was not reachable honestly; instead we added clinically sharper secondary tasks (e.g. moderate Parkinson's 0.96).

6. Teammate plan 1 (rigorous AD/FTD experiments)

• **3-class AD / FTD / healthy**: accuracy 0.65, macro-F1 0.62 - FTD is mostly confused with AD.

• **DT-GCN audit** (paper claims 82%): their code picks the best epoch on the test fold. Same model with honest selection: 0.52 accuracy; strict last-epoch: 0.38.

• **Graph ablation** (6 graph types × GCN/GraphSAGE/GATv2): a model with **no edges** scores the same as every graph (F1 0.48-0.55). Edges add nothing measurable.

• Re-ran Tigga et al.: edges inert; their leaky protocol 0.48 vs honest 0.44.

7. Focus on top 4 and BrainLat

• Narrowed to AD, FTD, depression, schizophrenia. Downloaded BrainLat (Synapse).

• Found that BrainLat patients and healthy people come from different file types (4× amplitude gap) - patient-vs-healthy scores of 1.00 were fake. Reviewed how others used BrainLat: careful studies get ~0.68-0.70.

• AD vs FTD on BrainLat is clean (same file type): 0.74 within BrainLat; 0.70 / 0.66 when trained on one dataset and tested on the other.

8. Teammate plan 2 (transformers, master model)

• Built the 'new' components from the teammate table: Graph Transformer, attention-MIL, GroupDRO, LaBraM fine-tuning - all trained as one **master model per disease** on all hospitals mixed.

• Result: Graph Transformer ties the simple models, LaBraM fine-tuning is mixed, attention-MIL and GroupDRO are worse. Weighting hospitals equally changed nothing. **AD vs FTD improved clearly** with the master model.

• Dropped leave-one-hospital-out and UNM.

9. Preprocessing review

• Literature: ICA / ASR artifact removal and reference changes do not improve classification (one study on our own Thessaloniki data).

• Tested three cheap changes. **Kept:** windows spread across the whole recording (median over windows) and 1/f-corrected band power - small but consistent gains (e.g. Warsaw 0.72 → 0.82, Mumtaz 0.92 → 0.96). **Dropped:** wPLI connectivity (worse).

• Caught a second confound: Riemannian covariances kept BrainLat's amplitude gap (fake 0.99). Dividing each covariance by its trace fixed it (0.70, in line with careful prior work) and even helped Thessaloniki FTD.

10. Final frozen system

• 5 members fixed before running: logistic regression, scale-free Riemannian, GATv2, Graph Transformer, fine-tuned LaBraM - all on scale-free inputs. Mean-rank ensemble, cross-fitted calibration, 'unsure' option.

• Beats or matches every earlier result (see 'Final system results').

11. Explainability and a last check

• Head maps per disease from within-hospital effect sizes, cross-checked with the model's weights.

• Found that Mumtaz's healthy group has ~55× more 44 Hz power than its depressed group (a filter difference). Using only 1-25 Hz, depression still scores 0.95, so the result stands with a caveat.

What worked and what did not

What we tried	Result	Verdict
Ensemble v2 (6 models voting)	Best within-dataset scores	Worked
Logistic regression / Riemannian / rich markers	Match or beat the graph networks	Worked
Master model per disease (all hospitals mixed)	Big gain for AD vs FTD; small losses elsewhere	Partly
Honest audits of published models (DT-GCN, Tigga)	Showed their scores rely on leaky testing	Worked
GATv2 / GCN / MSBP-GNN	All similar, none beats simple models	Partly
Graph edges (6 graph types)	No better than no edges	Did not
GATv2-LSTM (time sequence)	No gain	Did not
Graph Transformer	Ties the simple models	Partly
LaBraM pretrained transformer (frozen / fine-tuned)	Useful ensemble member, not a winner	Partly
Attention-MIL, GroupDRO	Worse on every task	Did not
ComBat, Euclidean alignment, DANN (hospital fixes)	Unseen hospital still near chance	Did not
Hospital-balanced training	No consistent change	Did not
Stricter depression cut-off (UNM)	No improvement; dataset dropped	Did not
3-class AD / FTD / healthy	65% accuracy; FTD confused with AD	Partly
Preprocessing v2: spread-out windows + 1/f-corrected power	Small consistent gains	Worked
wPLI connectivity instead of coherence	Worse	Did not
Scale-free (trace-normalised) Riemannian	Removed BrainLat fake 0.99; helped FTD	Worked
Final frozen ensemble + calibration + 'unsure'	Best numbers; 84-97% accuracy on confident half	Worked

Why the fancy models do not win

- **Too few people.** 28-130 people per dataset. A transformer with thousands of parameters memorises individuals; a simple model cannot.
- **Hospital fingerprints are stronger than disease.** Any flexible model happily learns the hospital.
- **The information is mostly in band power** (how much slow vs fast activity). Graph structure and time order add little on top - the ablation proved it for edges.

Final system results

One frozen system per disease (members fixed before running; master-model splits; every person tested by models that never saw them). AUC with 95% confidence interval. Results file: results/final_system.json.

Task	Hospital	AUC	95% CI	Before (master v1)
AD vs healthy	BrainLat*	**0.75**	0.62-0.86	0.74
	Thessaloniki	**0.94**	0.87-0.99	0.91
FTD vs healthy	BrainLat*	**0.64**	0.49-0.79	0.65
	Thessaloniki	**0.89**	0.78-0.97	0.81
AD vs FTD	BrainLat	**0.86**	0.75-0.95	0.86
	Thessaloniki	**0.71**	0.57-0.84	0.68
Depression	Mumtaz†	**0.97**	0.91-1.00	-
Schizophrenia	ASEEG	**0.84**	0.76-0.91	0.81
	Semmelweis	**0.67**	0.53-0.79	0.67
	Warsaw	**0.82**	0.63-0.96	0.67

* BrainLat patient-vs-healthy: scale-free inputs, but still carries the file-type caveat. † Mumtaz: 0.95 when restricted to 1-25 Hz (filter caveat).

Answer only when confident

The system can say 'unsure' for the people it is least confident about. Accuracy on the people it does answer:

Task	all	90%	80%	70%	60%	50%
AD vs healthy	77%	80%	82%	88%	89%	91%
FTD vs healthy	73%	72%	73%	74%	79%	79%
AD vs FTD	68%	70%	71%	73%	76%	82%
Depression	93%	94%	93%	95%	97%	97%
Schizophrenia	70%	72%	73%	76%	79%	84%

Every member, per hospital

Model	AD BrainLat	AD Thessaloniki	FTD BrainLat	FTD Thessaloniki	AD/FTD BrainLat	AD/FTD Thessaloniki
Logistic regression	0.72	0.92	0.69	0.79	0.86	0.63
Riemannian (scale-free)	0.70	0.88	0.65	0.86	0.76	0.74
GATv2	0.69	0.94	0.60	0.84	0.79	0.71
Graph Transformer	0.72	0.91	0.60	0.91	0.76	0.67
LaBraM fine-tuned	0.75	0.89	0.54	0.77	0.71	0.60
Ensemble	**0.75**	**0.94**	**0.64**	**0.89**	**0.86**	**0.71**
Model	**Dep Mumtaz**	**SZ ASEEG**	**SZ Semmelweis**	**SZ Warsaw**		
Logistic regression	0.95	0.77	0.68	0.84		
Riemannian (scale-free)	0.91	0.81	0.59	0.87		
GATv2	0.97	0.80	0.53	0.78		
Graph Transformer	0.95	0.70	0.51	0.70		
LaBraM fine-tuned	0.96	0.79	0.68	0.56		
Ensemble	**0.97**	**0.84**	**0.67**	**0.82**		

Earlier master model (before preprocessing v2), every model

Results file: results/pooled_deep.json. All models used identical splits; differences under ~0.10 are within noise.

AD vs control

Model	Thessaloniki	BrainLat	All
Logistic regression	0.92	0.73	0.83
GATv2 (graph)	0.89	0.67	0.79
Graph Transformer	0.89	0.74	0.82
Attention-MIL	0.82	0.64	0.74
Attention-MIL + GroupDRO	0.83	0.69	0.76
LaBraM fine-tuned	0.88	0.72	0.81
Ensemble (all 6)	0.91	0.74	0.83

FTD vs control

Model	Thessaloniki	BrainLat	All
Logistic regression	0.76	0.69	0.72
GATv2 (graph)	0.79	0.69	0.73
Graph Transformer	0.81	0.58	0.70
Attention-MIL	0.67	0.62	0.65
Attention-MIL + GroupDRO	0.73	0.60	0.67
LaBraM fine-tuned	0.73	0.52	0.64
Ensemble (all 6)	0.81	0.65	0.73

AD vs FTD

Model	Thessaloniki	BrainLat	All
Logistic regression	0.62	0.86	0.74
GATv2 (graph)	0.67	0.79	0.72
Graph Transformer	0.69	0.83	0.75
Attention-MIL	0.51	0.66	0.58
Attention-MIL + GroupDRO	0.57	0.71	0.64
LaBraM fine-tuned	0.62	0.68	0.65
Ensemble (all 6)	0.68	0.86	0.76

Schizophrenia

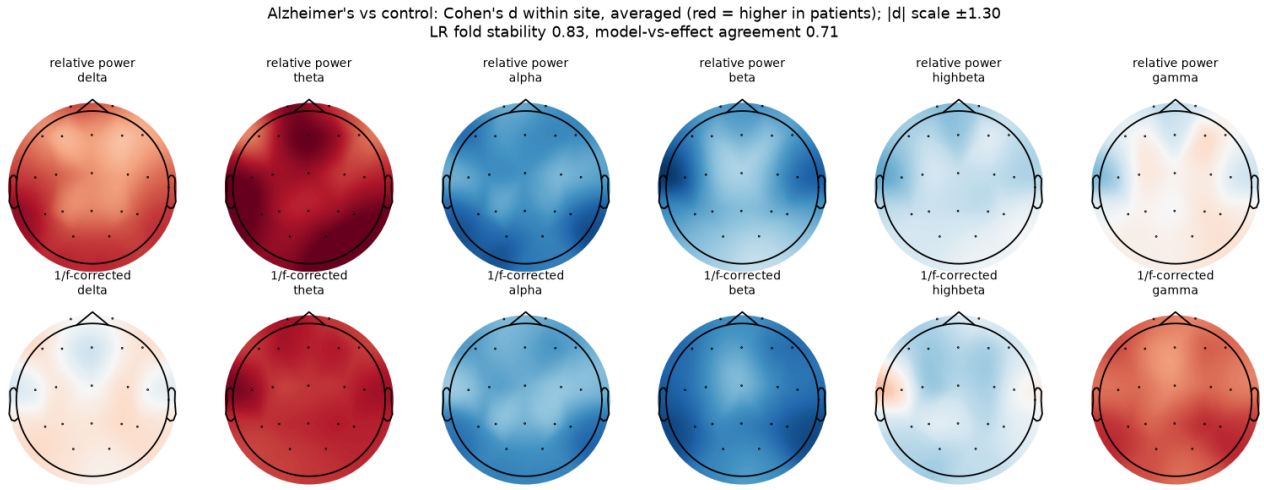
Model	ASEEG	Warsaw	Semmelweis	All
Logistic regression	0.75	0.72	0.69	0.72
GATv2 (graph)	0.81	0.79	0.63	0.75
Graph Transformer	0.78	0.70	0.56	0.70
Attention-MIL	0.73	0.63	0.60	0.68
Attention-MIL + GroupDRO	0.70	0.52	0.61	0.65
LaBraM fine-tuned	0.79	0.44	0.70	0.72
Ensemble (all 6)	0.81	0.67	0.67	0.74

What the system looks at

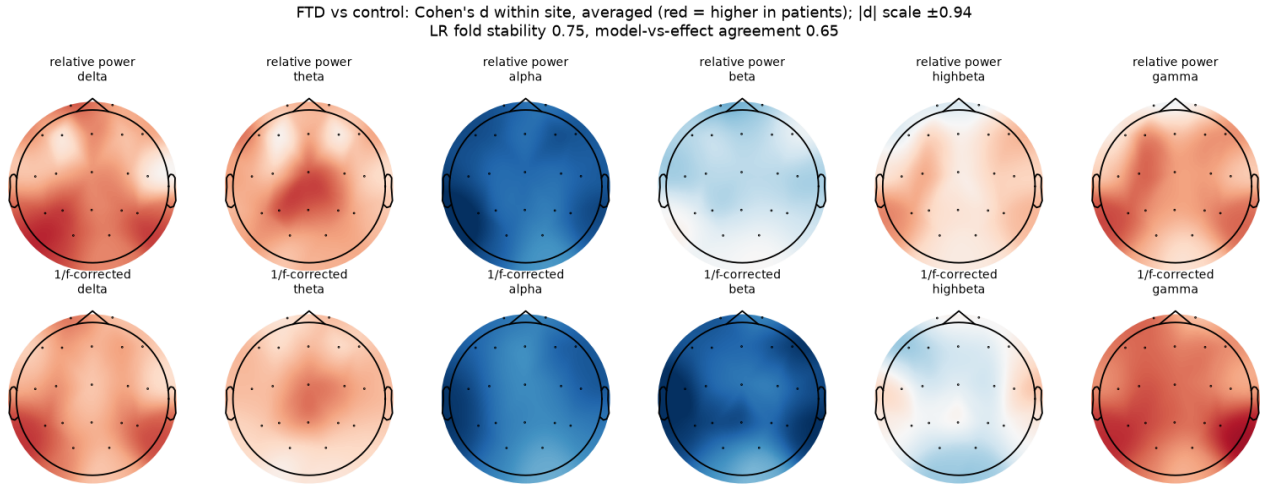
Maps show, for each frequency band, where patients differ from healthy people (red = higher in patients, blue = lower), measured inside each hospital and then averaged. They are cross-checked against the logistic-regression member's weights: 'stability' = do the 5 training folds agree, 'agreement' = does the model use what the data shows.

- **Alzheimer's:** more slow theta everywhere, less alpha/beta - the classic 'EEG slowing' of dementia. Same direction at both hospitals.
- **FTD:** less beta and alpha, strongest over temporal and frontal electrodes.
- **Alzheimer's vs FTD:** Alzheimer's shows more theta than FTD.
- **Schizophrenia:** right-temporal (T4) delta up, alpha down; weaker effects (d ≈ 0.5).
- **Depression (Mumtaz):** dominated by a 44 Hz filter difference between groups - see caveats; the 1-25 Hz pattern alone still separates the groups.

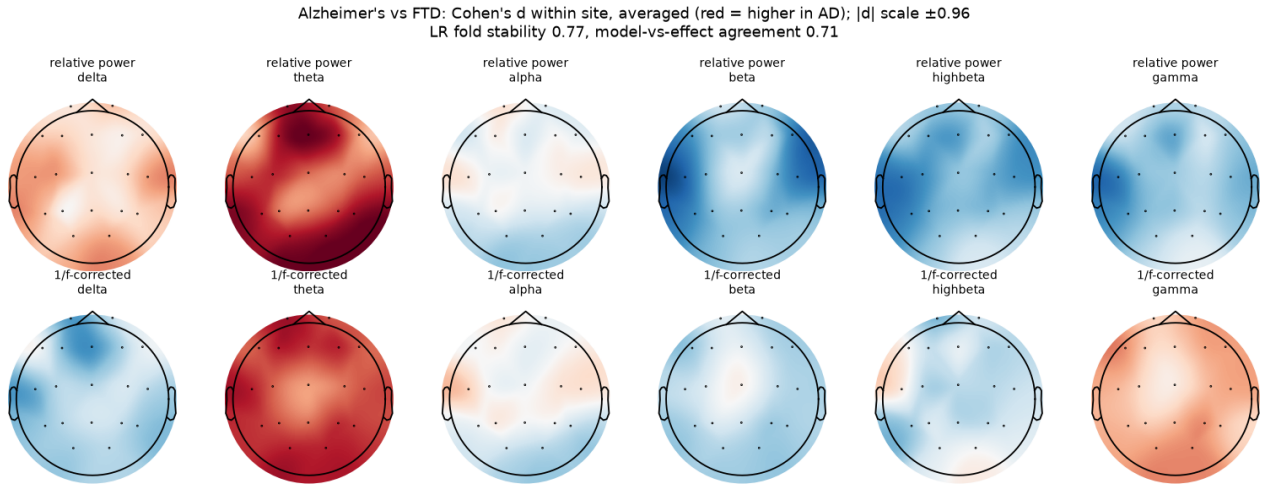
Alzheimer's vs healthy



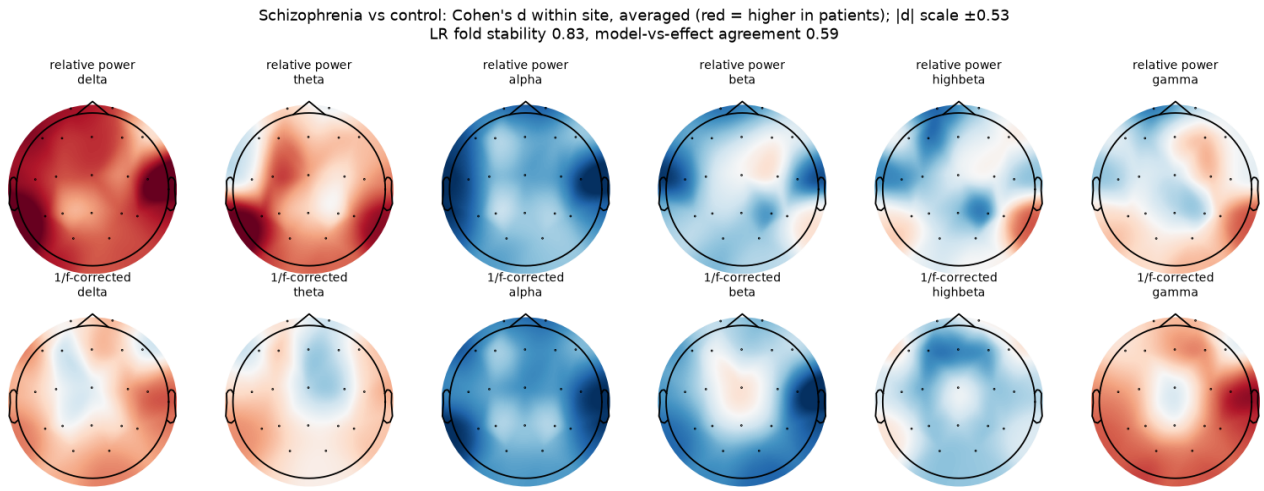
FTD vs healthy



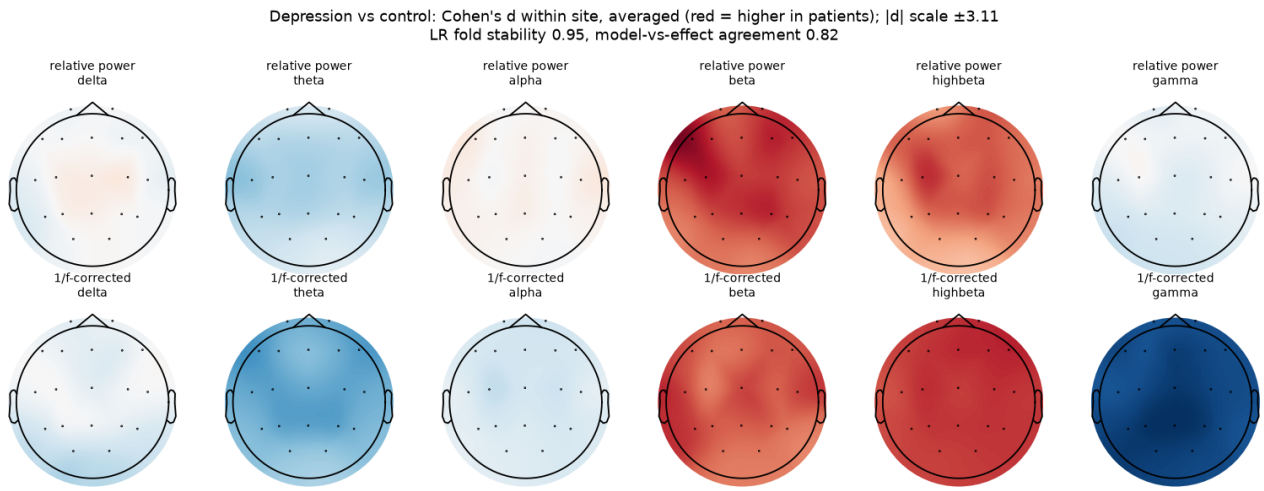
Alzheimer's vs FTD



Schizophrenia vs healthy



Depression vs healthy (Mumtaz)



Status of the teammate's architecture table

Component	Status	Verdict
GATv2, GCN, MSBP-GNN, GATv2+LSTM	Built and benchmarked	Done
Riemannian, LaBraM, ComBat, Euclidean alignment, DANN	Built and benchmarked	Done
Graph Transformer	Built (master model) - ties simple models	Done
LaBraM deeper fine-tuning	Built (last 2 blocks) - mixed	Done
Attention-based MIL	Built - worse	Done
GroupDRO	Built - worse	Done
Uncertainty-aware prediction	Built: calibrated probabilities + 'unsure' option	Done
Spectral / hierarchical temporal transformer, cross-attention	Skipped: need far more data; LaBraM covers the idea	Dropped
Supervised contrastive learning	Skipped for now: little to contrast with ~100 people	Dropped

Honest caveats to keep in the report

- **Depression is single-site** (Mumtaz, 58 people). 0.97 cannot be claimed to generalise. Option: request access to MODMA (clinically diagnosed depression, needs a signed request).
- **Mumtaz has a filter difference between groups** (controls ~55× more 44 Hz power). Restricted to 1-25 Hz the score is 0.95, so quote 0.95 with this caveat, not 0.97.
- **BrainLat patient-vs-healthy** is confounded by recording file type. We remove the amplitude gap, but other differences may remain. AD vs FTD on BrainLat is clean.
- **Small datasets** mean wide confidence intervals; many model differences are noise.
- **Generalising to a brand-new hospital is not solved** - we tried and documented it.

What is left

#	Task	Status
1	Experiments, final system, calibration, explainability	Done
2	Report, results page, model registry	Updated with these results
3	GitHub	Merged into master; code reorganised into the eeg/ package
4	Optional: 3-class AD/FTD/healthy on the final system	Not started (~30 min)
5	Optional: second depression dataset (MODMA)	Needs a signed access request - team decision

Decisions for the team

- Is single-site depression acceptable, or do we request MODMA?
- Agree the headline framing: an honest benchmark showing that simple models match GNNs/transformers on current public EEG, plus audits of leaky published results - rather than chasing one big number.

Code: github.com/AbdulMalik1287/MiniProject-I (master). Start at the README; the final system is eeg/experiments/final_system.py, its explanation eeg/experiments/explain_final.py, and every model is listed in docs/MODEL_REGISTRY.md.